GRIDPRINT analysis report

Verdict

VERIFIED

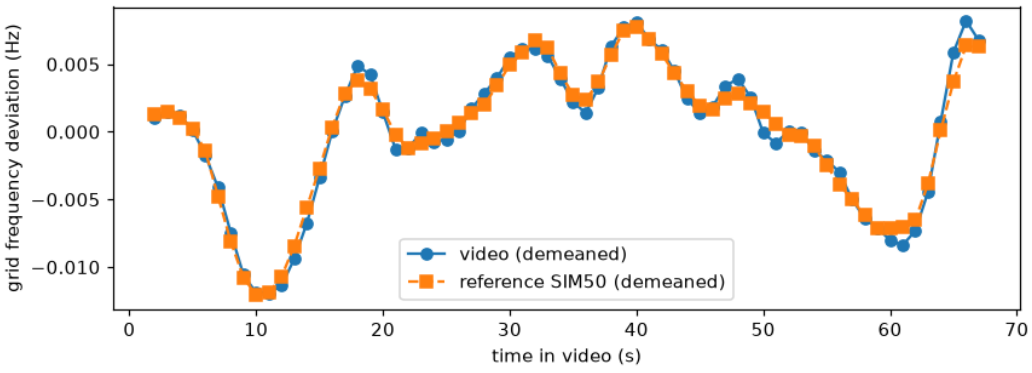
A grid signal is present and matches the reference log at one time. No significant edit evidence was found.

Report state: FINAL

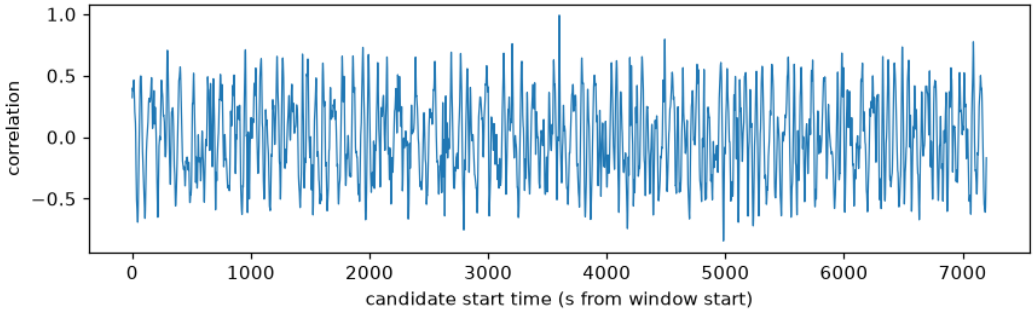
Summary

Estimated recording start (UTC)	2026-01-20T09:59:59.973488+00:00 ± 1.5 s
Grid zone	SIM50
Reasons	• `READOUT_FROM_SIDECAR` — Readout time came from the Capture sidecar. • `MATCH_OK` — The track matches the reference log at one time.
Match	peak correlation 0.991, next peak 0.795, track length 69 s

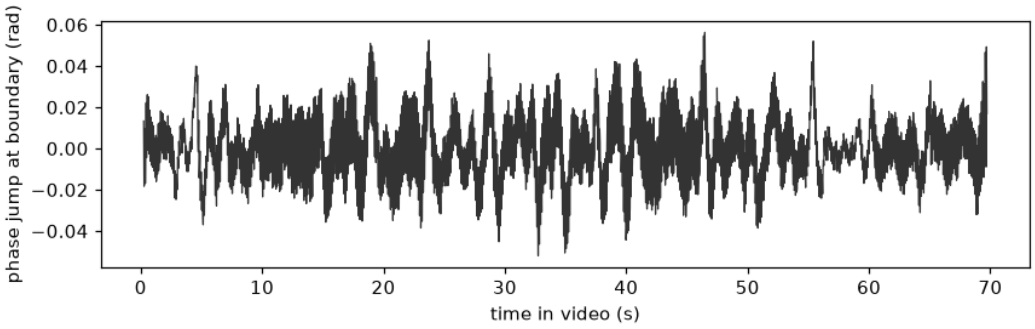
Key plots



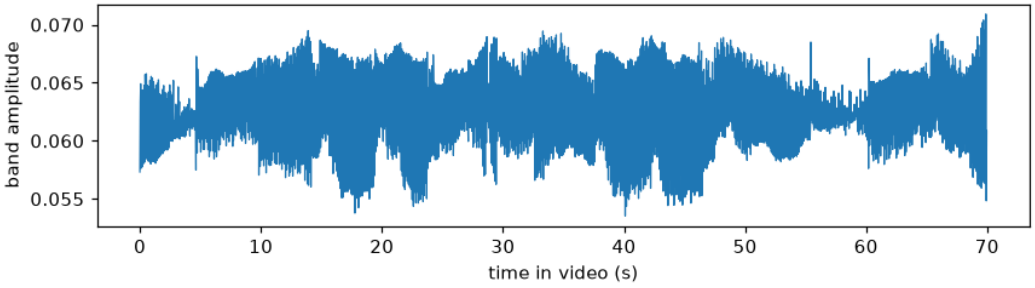
Measured grid frequency variation and matched reference.



Correlation against candidate start times.



Phase jump at frame boundaries. Solid line: edit candidate. Dotted line: timestamp gap.



Flicker band amplitude per frame (fractional modulation).

Limitations

- GRIDPRINT reads the grid frequency from the flicker of mains-powered lights. It is not a deepfake classifier and it is not cryptographic proof.
- A missing signal is never evidence that a video is fake. Many phones hide the flicker ("anti-banding"), and daylight or battery lights do not flicker.
- A time estimate is possible only inside the time range of the reference log for the zone.

- Location output is limited to the grid zone. GRIDPRINT gives no finer location.
- Thresholds come from calibration data. Results outside the calibrated conditions carry more risk of error.
- A CONTRADICTED verdict needs approval by a human reviewer before the report is final.

Provenance

File SHA-256	`b77ced2d949742114aa1c67988026396ef14c72cfd7221ff30663d595d7c7779`
Engine version	0.1.0
Config hash	`8f439372b865ee8e`
Thresholds hash	`0161b96f77d3e925`
Job	`aws4-capture-verified-041747`

Appendix

Continuity events

None.

Quality metrics

axis	rows+
band_snr_db	6.614609559814535
bitrate	475816
bits_per_pixel	0.2065173611111111
camera_motion_px	0.0332642155899567
codec	h264
dft_peak_hz	100.02936157407733
estimators_agree	True
foreground_fraction	0.052797619047619045

frames	2100
height	240
jitter_p50_ms	3.3302427482340136e-07
jitter_p95_ms	3.3385298969790256e-07
motion_level_px	0.1115635689380337
nominal_flicker_hz	100.0
opencv_frame_count	2100
phase_mean_flicker_hz	100.02602150557438
readout_s	0.025
readout_source	READOUT_FROM_SIDECAR
rotation_metadata_deg	0
segmentation	ok
static_fraction	1.0
timestamp_source	sensor
timing_assumption	sensor_timestamp_is_first_row_exposure_start
track_id	t1
track_samples	66
transfer	bt709
verdict_rule	S13.4
width	320

Stage timings (s)

S1_ingest	0.2958
S2_linear	2.1547
S3_readout	0.0
S4_motion	6.4506
S5_segment	0.9503
S7_rowsignal	1.3239
S6_patches	1.5129

S8_phase	0.7567
S10_match	0.7072
S11_continuity	0.0009